

When Does Structure Help a Magnitude-Only Channel Estimator? Aperture, Effective Rank, and a Prospective Test

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Abstract—Rydberg atomic receivers sense a radio field through an optical transition and report its amplitude alone, so the phase is lost at the point of measurement. This paper asks when it helps to constrain the channel in such a receiver. On a uniform linear array the channel across antennas is a sum of a few sinusoids, which is a strong constraint but a bounded one: the array size fixes how much structure it can carry. Two properties of the receiver settle the question in advance. Each amplitude reading constrains the channel in one direction only, and readings at different antennas are independent of one another, so extra antennas do not improve accuracy per antenna unless the estimator exploits how the channel varies across them. The constraint should therefore help when the channel is simple relative to the array’s capacity and hurt otherwise, which is what we measure: averaged over twelve operating points per array size, it gains +2.93 dB at 32 antennas and loses 0.25 dB at 8, while the unconstrained baseline varies by at most 0.163 dB. The loss appears only in summed squared error; under a per-trial median the same cells give +0.225 dB, because the estimator declines the constraint in 42% of trials. The governing quantity is not the path count but the fraction of the array’s capacity the channel occupies, and it predicted the gain on two unseen channel distributions to within 0.13 and 0.315 dB. Results are simulation only, and the bounds computed here govern unbiased estimators while every estimator studied is biased.

Index Terms—Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, constrained Cramér–Rao bound, effective rank.

I. INTRODUCTION

SIXTH-GENERATION systems are expected to operate at higher carrier frequencies and over wider bandwidths than their predecessors, and both trends press on the receiver: higher frequencies mean weaker incident fields, and wider bandwidths mean the front end must stay linear and calibrated across more spectrum. Conventional receivers answer with metallic antennas and electronic amplification, whose sensitivity is bounded by thermal noise in the antenna and the first amplifier stage, and which must be recalibrated band by band. Quantum sensing has emerged as one response, and among its candidates the Rydberg atomic receiver has drawn the most attention for communications [1], [2]. Such a receiver replaces the antenna with a glass cell of alkali vapour whose atoms are laser-excited into high-lying Rydberg states. An incident radio field shifts the atomic energy levels, and a probe laser crossing the cell carries that shift out as a

change in optical transmission, so sensitivity is set by atomic physics rather than circuit noise, and the response is retuned by changing laser frequencies rather than by rebuilding hardware. One consequence organises everything that follows: the optical readout responds to the *amplitude* of the radio field, so the phase is destroyed at the point of measurement and never appears at the output.

Channel estimation is therefore not a linear inverse problem: recovering a complex channel from amplitude-only measurements is phase retrieval, non-convex and without a closed form, made tractable here by a known reference field that converts relative phase into observable amplitude. Existing estimators for this receiver improve the solver — alternating projection [3], gradient descent with a rank projection [5], or an unrolled network [4] — but none addresses when constraining the channel is worth doing at all. A constraint is reported as helpful because it helped at the array size and channel richness simulated, with no account of when it would stop helping or start hurting. That gap is the subject of this paper: under what conditions does imposing structure actually improve a magnitude-only estimator, and how could a designer tell in advance?

Contributions.

- We show that the information carried by an amplitude-only measurement is block diagonal across receive elements under the unconstrained parameterisation of the channel, so that normalised per-element information does not grow with array size, and we identify the consequence: aperture is usable only by an estimator that exploits how the channel varies across elements.
- We derive the capacity of the exponential-sum structure on a uniform linear array, obtain from it a prediction that a rank constraint must fail and then do harm at small aperture, and test that prediction against measurement.
- We replace the path count by the effective rank of the channel relative to that capacity, and separate what the constraint contributes from what the estimator’s decision rule contributes.
- We freeze the resulting relation, commit numerical predictions, and then evaluate them on two channel distributions that played no part in constructing it.

II. SYSTEM MODEL AND ASSUMPTIONS

An N -element uniform linear array (ULA) of vapour cells serves K single-antenna users over P pilot symbols, as in

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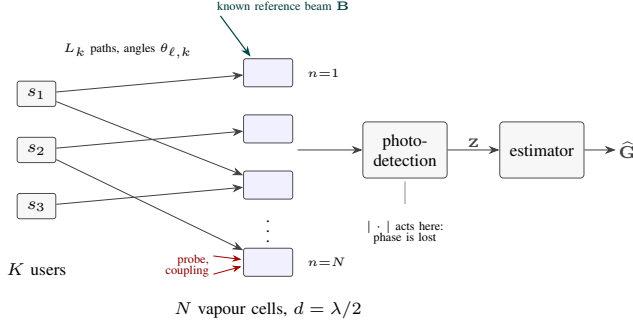


Fig. 1. Uplink of the atomic receiver. K single-antenna users transmit pilots $\mathbf{S}$ over a geometric multipath channel to N vapour cells arranged as a uniform linear array. Each cell is interrogated by probe and coupling lasers and illuminated by a known reference beam $\mathbf{B}$. Photodetection reports the magnitude of the total field, which is where the phase is lost, and the estimator sees only $\mathbf{Z}$.

Fig. 1. With $\mathbf{G} \in \mathbb{C}^{N \times K}$ the channel, $\mathbf{S}$ the pilots, $\mathbf{B}$ the known reference beam and $\mathbf{W}$ the noise, the receiver observes

$$\mathbf{Z} = |\mathbf{G}\mathbf{S} + \mathbf{B} + \mathbf{W}|, \quad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2), \quad (1)$$

with $|\cdot|$ elementwise. The modulus is applied after the noise is added, which is what makes (1) impossible to linearise without approximation. Each user's channel is a sum of L_k plane waves on the array manifold,

$$g_k[n] = \sum_{\ell=1}^{L_k} \alpha_{\ell,k} e^{-j(n-1)\psi_{\ell,k}}, \quad \psi_{\ell,k} = \pi \sin \theta_{\ell,k}, \quad (2)$$

where ψ follows from half-wavelength element spacing. The reference-to-signal ratio (RSR) is the reference power divided by the *single-user* received power, and the signal-to-noise ratio (SNR) is the single-user received power divided by σ^2 ; both are defined at the array and are design parameters rather than measured quantities.

Assumptions. The model above rests on the following, stated so that each can be attacked separately.

- Sparse geometric multipath, with L_k small relative to N .
- Path gains $\alpha_{\ell,k}$ i.i.d. circularly symmetric complex Gaussian, with equal expected power across paths.
- Angles of arrival i.i.d. uniform on $(-90^\circ, 90^\circ)$, with $L_k \sim \mathcal{U}\{3, \dots, 7\}$ [5].
- Narrowband and frequency flat: no delay spread, so path delays across the array act as pure carrier phase.
- Equal user power, so near-far effects are untested here.
- Pilots drawn once per trial and shared bit-identically by every estimator compared.
- Noise added before the modulus, as in (1).
- Reference beam known at the receiver in both amplitude and phase.

III. PROPOSED ESTIMATOR

A. Structure and its capacity

Substituting (2) into the $(N - p) \times (p + 1)$ Hankel lifting $\mathcal{H}_p$ separates the exponential in its two indices, so each path contributes one rank-one outer product and $\text{rank } \mathcal{H}_p(\mathbf{g}_k) \leq L_k$. The lifting is therefore rank deficient whenever the channel

has few paths [11], [12]. Because $\mathcal{H}_p$ has dimensions $(N - p) \times (p + 1)$, the largest rank it can carry is

$$r_{\max}(N) = \max_p \min(N - p, p + 1) = \lceil N/2 \rceil, \quad (3)$$

and if $L_k \geq r_{\max}(N)$ the constraint says nothing. This yields a prediction available before any simulation: at $N = 8$ the capacity is 4 while the simulated path counts run from 3 to 7, so the constraint is frequently vacuous and, when it does bite, removes genuine components. A rank constraint should therefore help at large aperture and harm at small.

The user count cancels: the structural model uses $3 \sum_k L_k$ parameters against $2NK$ unstructured, a ratio independent of K , so the gain is expected to be invariant to the number of users at fixed pilot adequacy.

B. The estimator

After every exact expectation-maximization Gerchberg-Saxton (EM-GS) update, the proposed estimator applies a Cadzow projection $\mathcal{P}_r(\mathbf{g}) = \mathcal{H}_p^\dagger(\Pi_r(\mathcal{H}_p(\mathbf{g})))$, in which Π_r truncates to the r largest singular values and $\mathcal{H}_p^\dagger$ averages along antidiagonals. One application is *not* idempotent: it is a single step of an alternating projection between the set of rank- r matrices and the set of Hankel matrices, so the number of sweeps is part of the operator's definition and not a tuning detail. With the projection disabled the estimator reproduces EM-GS to $\max |\text{diff}| = 0$, so the projection is provably the only difference between the two arms compared throughout.

C. Choosing the order without an oracle

The rank r is not known and is not supplied. A fraction $\nu = 0.3$ of the pilot columns is held out, each candidate $r \in \{1, \dots, r_{\max}\}$ is scored by the magnitude residual on those held-out columns, and the minimiser is used. An in-sample criterion cannot serve: constraining $\mathbf{G}$ can only worsen the fit to the pilots used for fitting, so the in-sample residual always prefers the largest candidate. The true L_k never enters, so the rule uses no oracle. When the selector returns $\hat{r} \geq r_{\max}$ the projection is the identity and the estimator *abstains*, falling back exactly to EM-GS. This is designed behaviour rather than a failure mode, and it turns out to carry much of the method's protection against a badly chosen constraint.

Subspace methods such as root-MUSIC and ESPRIT would recover $\{\psi_{\ell,k}\}$ directly, but they need the signal subspace, which requires either multiple snapshots or the complex field. With magnitude-only single-snapshot data that subspace is identifiable only after phase retrieval has succeeded, so we regularise toward the exponential-sum manifold rather than parameterising it.

Structured low-rank Hankel recovery for sums of exponentials is well developed for *linear* measurements [7], [8], [9], as are gridless line-spectrum methods under a separation condition [10]. Neither applies here: the measurement is a modulus, so no convex relaxation is available and the projection can act only as a regulariser inside a non-convex iteration.

IV. PERFORMANCE REFERENCES

A. Information in an amplitude measurement

Each observation is Rician and depends on the noiseless field only through its modulus, so it constrains the channel along a single direction: the information it contributes is rank one [6], and for this observation model the exact Rician form is derived in [1]. Crediting both quadratures instead yields a reference too low by 0.17–4.64 dB across our 36 operating points, and at $K = 3$, $P = 3$ it reports full rank where the per-element problem is in fact unidentifiable. The existing bound for this receiver [5] is computed from a linearised model; the one used here keeps the exact Rician form.

The consequence matters more here than the form. Because the n th row of **GS** enters only the n th row of the measurement, the information matrix is block diagonal across receive elements, so the per-element problem is unchanged by adding elements and normalised per-element information does not grow with N . An unstructured estimator therefore gains nothing from a larger array, and aperture is available only to an estimator that couples elements — which is what (3) supplies.

B. A reference that accounts for the structure

The ordinary Cramér–Rao bound describes an estimator that treats every channel entry as a free unknown. The proposed estimator instead uses the fact that the channel is a sum of a few sinusoids, and the constrained bound is the corresponding limit for estimators that do; the gap between the two is what the structure is worth in principle. Under the geometric model the parameters number about 45 and, unlike the $2NK$ unstructured coordinates, that count does not grow with N . The textbook constrained form is unusable because the Jacobian **D** is rank deficient, its mean numerical rank at $N = 8$ being 40.4 against ≈ 44.7 parameters, so we use the tangent-space form [14], [15]

$$\text{CCRB} = \mathbf{U}(\mathbf{U}^T \mathbf{J} \mathbf{U})^{-1} \mathbf{U}^T, \quad (4)$$

with **U** an orthonormal basis for the directions the geometric model can reach, which is well defined when **D** is not full rank.

Both references constrain the covariance of *unbiased* estimators. Every estimator here runs to a fixed iteration budget with a shrinking update, and the proposed one additionally truncates rank, so none is unbiased and neither reference is a bound on what we measure. We report them as reference curves, with the constrained one primary because it is matched to the estimator’s model class, and the unconstrained one secondary because it describes the baseline only.

V. NUMERICAL RESULTS

A. Experimental setup

Channels follow (2) with the parameters of Section II. Two generators are used: TB in Table I draws a fresh channel per trial at a fixed SNR per operating point, and TD draws the SNR itself per trial uniformly over $[-10, 20]$ dB and bins afterwards, which is what its $\mathcal{U}$ entry denotes. SNR and RSR

TABLE I
THE FOUR MEASUREMENT CONFIGURATIONS. P IS ITSELF SWEEPED IN CONFIGURATION C’S USER-COUNT CELLS (13–27) AND PILOT-COUNT CELLS (10–35).

Config.	T	order search	P	RSR (dB)	Generator	SNR (dB)
A	100	25	10, 30	12	TB	$\{-5 \dots 20\}$
B	50	20	30	12	TB	$\{-5 \dots 20\}$
C	100	25	20	10	TD	$\mathcal{U}[-10, 20]$
D	50	20	30	12	TB	5

are as defined in Section II, both at the array, and noise enters before the modulus.

Unless a sweep varies them, $N = 32$ elements serve $K = 3$ users at RSR 12 dB with 300 to 400 trials per point, the pilot count stated per result. Trials are paired: both estimators receive bit-identical channels, pilots, reference and noise, so a per-trial difference carries no realisation noise. The reported statistic is the paired per-trial median within an SNR bin with bootstrap 95% intervals over trials; where a curve is compared against a reference we use the ratio of summed errors, and both are reported wherever they disagree. With the projection disabled the estimator reproduces the baseline bit for bit, so the projection is the only difference between the arms throughout.

Four configurations appear because the results were produced over time at different operating points, and collapsing them would misrepresent what was measured. All share the estimator and the order rule, differing in T , the order-search budget, the pilot count, the RSR and the generator. A and B differ only in T and the search budget, and at the one cell measured both ways the difference is 0.005 dB. Every number below carries its configuration label.

B. Performance against the references

In Fig. 2(a) the NMSE of both estimators is plotted against SNR at configuration A with $N = 32$ and $P = 30$, together with the two references. The proposed estimator improves on the baseline by +2.46 to +3.05 dB across the sweep, with a six-point mean of +2.68 dB and a ratio-of-sums improvement of +2.33 to +2.96 dB, the projection being active in 100% of trials at this cell.

The baseline tracks the unconstrained reference to within 0.143 dB for $\text{SNR} \geq 5$ dB, so that reference is informative rather than vacuous. The constrained reference lies 7.05–7.11 dB below it here, and that gap is what the structural model is worth in principle; the proposed estimator recovers part of it, remaining above the constrained reference at all six points by 4.39–4.93 dB.

C. Aperture, and which statistic reverses

The capacity argument predicts that the gain turns negative at small aperture. Fig. 2(b) plots Δ_{HS} against $N \in \{8, 16, 32\}$ at configuration A, each point the mean over 12 operating points (2 pilot counts $\times$ 6 SNR values), so that both panels share one operator. The two pooling rules agree except at $N = 8$, so both are reported:

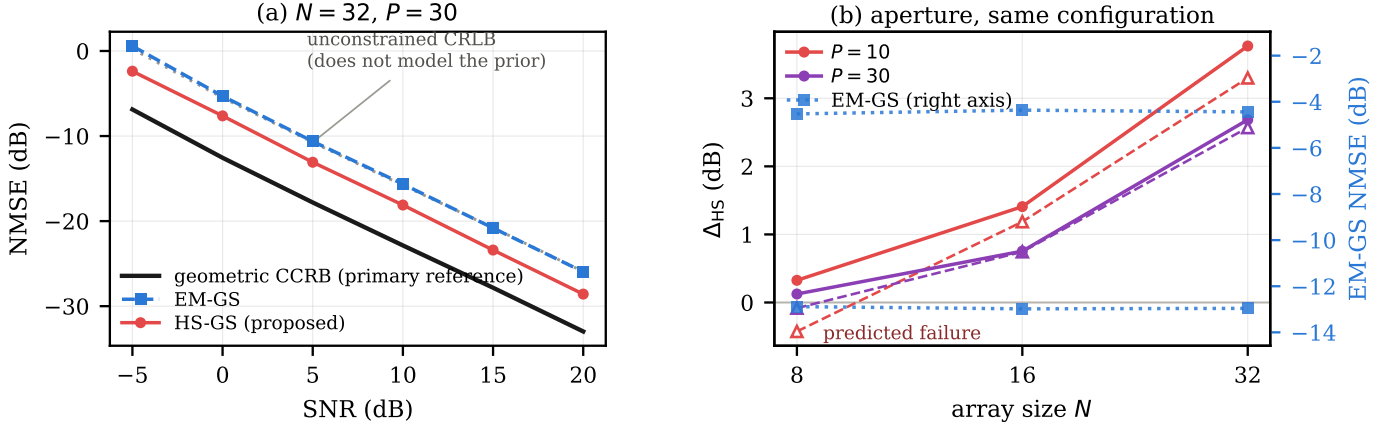


Fig. 2. Both panels are configuration A ($n_{cz} = 4, T = 100$, order search 25, RSR 12 dB, $K = 3$, 400 paired trials per point), so they share one operator. (a) NMSE against SNR at $N = 32, P = 30$; the unconstrained reference is faint because it does not model the constraint, and neither reference governs a biased estimator. (b) Δ_{HS} against N at both pilot counts, each point the mean over six SNR values. Filled markers are the paired per-trial median, open markers the ratio of summed errors; the two disagree at $N = 8$. The baseline's own NMSE is overlaid on the right-hand axis.

N	ratio of sums	paired median	baseline NMSE
8	-0.254	+0.225	-8.705
16	+0.966	+1.081	-8.672
32	+2.934	+3.226	-8.700

The baseline varies by at most 0.163 dB over this fourfold change in aperture — 0.163 dB at $P = 10$ and 0.097 dB at $P = 30$ — against a structured gain moving by over 3 dB. Pooling the pilot counts first gives 0.033 dB, but they move in opposite directions, so we quote the larger figure. This is what the block-diagonal information structure requires of an estimator that does not couple elements.

The predicted reversal is a ratio-of-sums phenomenon. At $N = 8$ the projection is inactive in 41.8% of trials on average, rising from 12% at -5 dB to 59% at $P = 10$ and 79% at $P = 30$ at +20 dB. Where it is active it removes genuine components, and the ratio of summed errors turns over accordingly, from +1.272 dB at -5 dB to -2.613 dB at +20 dB at $P = 10$, truncation bias not shrinking with the noise while the variance term does.

The paired median does not reverse. At $N = 8$ it is +0.000 dB at nine of the twelve points, positive and significant at the other three, and negative and significant at none. Abstention makes most trials bit-identical to the baseline and pins the median at zero, while the active minority dominates a sum of squared errors. The prediction is therefore confirmed on the statistic a bound governs and invisible to the one we otherwise treat as primary.

The bands registered before this sweep ran were $[-0.35, +0.05]$, $[+0.60, +0.95]$ and $[+2.65, +3.00]$ dB. The values at $N = 8$ and $N = 32$ fall inside their bands and the one at $N = 16$ misses by 0.016 dB, so the pre-registration failed. The baseline spread was predicted to be at most 0.15 dB; it is 0.033 dB pooled, which is what the prediction named, and 0.163 dB at $P = 10$ alone, which would fail it. A failed prediction licenses no reporting other than the measurement.

D. Path count is the wrong coordinate

At configuration D, holding $N = 32$ and varying only the path count with L identical across users at 5 dB SNR, the gain decays monotonically from +7.04 dB at $L = 2$ to -0.12 dB at $L = 16 = r_{\max}$, while the baseline moves by 0.191 dB in total. The operative variable is therefore channel complexity relative to capacity, and aperture matters only because it sets that capacity.

E. Effective rank, and the constraint versus the estimator

Path count is still imperfect, because clustered or unequal-strength paths occupy fewer effective dimensions than their number suggests. We index instead by the Roy-Vetterli effective rank [13] of the lifting, $r_{\text{eff}} = \exp(-\sum_i p_i \ln p_i)$ with $p_i = \sigma_i / \sum_j \sigma_j$, normalised as $\rho = r_{\text{eff}} / r_{\max}$. A 16-path channel on a 32-element array is algebraically full rank yet has effective rank 8.73, so its lifting is compressible with nothing sparse about it.

Re-indexing the path-count sweep at configuration D moves the zero crossing from $L/r_{\max} = 0.90$ to $\rho = 0.518$, and the useful region becomes $\rho \lesssim 0.5$ — a statement about any channel rather than about one simulator's L values. In Fig. 3, Δ_{HS} is plotted against ρ for $N \in \{16, 32, 64\}$, with crossings 0.588, 0.518 and 0.544, a spread of 0.070 over a fourfold change in aperture. This comparison spans two configurations — the $N = 32$ value is configuration D and the other two are configuration C — and only the $N = 32$ crossing is bracketed by measurements on both sides. The other two are extrapolated from the positive side, because abstention prevents the proposed estimator from producing negative values there.

That separates the constraint from the estimator. Past the crossing at configuration C the paired per-trial median never becomes negative: at $\rho = 0.608$ and 0.735 for $N = 16$ it is +0.078 and +0.000 dB, and at $\rho = 0.596$ and 0.669 for $N = 64$ it is +0.044 and +0.018 dB. The reason is abstention — at $\rho = 0.735$ the selector returns $\hat{r} \geq r_{\max}$ in 38.8% of trials and the estimator is then exactly the baseline. Forcing the

projection active at that same cell, as a diagnostic that is not the proposed method, gives -0.075 dB with a confidence interval of $[-0.186, -0.026]$. The constraint does become harmful past the crossing, and abstention is what stops the typical trial from following it down. So $\Delta_{\text{HS}} \geq 0$ is a property of the selector under a median statistic only, and is neither evidence that the constraint is never harmful nor a guarantee about mean-square error.

The magnitude of the gain does not collapse onto the same axis. Over the window both arrays cover at configuration C, $N = 16$ lies above $N = 64$ by a mean of 0.493 dB and a maximum of 0.901 dB, one-signed throughout, and we have no account of this residual. The one candidate we tested — that the selector’s rank grid is coarser at small capacity — is refuted by a pencil sweep at fixed $N = 32$ and fixed ρ , which shows no monotone trend with the number of admissible ranks and tracks that count rather than the matrix shape.

Finally, ρ is a design-time coordinate and not a receiver-observable one, since r_{eff} is computed on the noiseless channel. The abstention rate is observable and monotone in ρ at fixed aperture, but confounded by capacity: at $N = 64$ a cell at $\rho = 0.501$ abstains in 0.0% of trials while at $N = 16$ a cell at $\rho = 0.371$ abstains in 4.0% , so no single threshold separates the useful region. An observable proxy therefore exists within an aperture but has not been established across apertures.

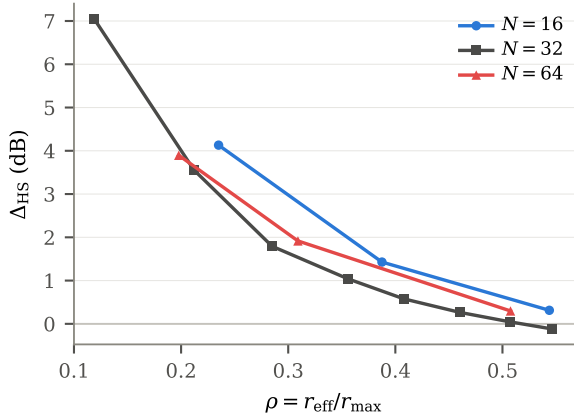


Fig. 3. Δ_{HS} against $\rho = r_{\text{eff}}/r_{\text{max}}$ for $N \in \{16, 32, 64\}$. The boundary location is stable across a fourfold change in aperture; the gain magnitude is not, and that residual is unexplained.

F. Prediction on unseen channel distributions

A relation fitted and evaluated on the same data proves little. We froze the ρ -to-gain relation obtained above, committed numerical predictions to version control, and only then ran two channel generators that had played no part in constructing it. The rule is piecewise-linear interpolation on a fixed eight-point table, with no regression and no spline, so it re-executes exactly. Both tests use configuration C’s operator and operating point with the generator replaced.

For a clustered Saleh–Valenzuela generator at $\rho = 0.331$ the relation predicted $+1.30$ dB, and the measurement over 276 trials gave $+1.173$ dB with a confidence interval of $[+1.017, +1.380]$, an error of 0.13 dB. For a second, independently specified configuration at $\rho = 0.400$ the prediction

was $+0.648$ dB with interval $[+0.367, +1.015]$, and the measurement over 266 trials gave $+0.963$ dB with a confidence interval of $[+0.808, +1.086]$, an error of 0.315 dB — inside the stated interval but on its upper edge.

These are cross-generator and not out-of-model tests: both generators still produce channels of the form (2), and only the distribution over $(L_k, \alpha_{\ell,k}, \theta_{\ell,k})$ changes. They show that ρ transfers across path distributions, not that it survives a violation of the ULA model.

G. Users, cost, and where the projection belongs

At configuration C and fixed pilot adequacy $P/2K = 3.33$, the gain varies by 0.094 dB across $K \in \{2, 3, 4\}$ for $\text{SNR} \geq 5$ dB, the invariance the parameter-ratio argument predicts. Pooled over all SNR the spread is 0.294 dB against a pre-registered falsifier of 0.3 dB, missed by 0.006 dB and reported rather than dropped.

Order selection accounts for 57.7 – 85.5% of runtime, so it, and not the projection, is where an implementation should look for savings.

Comparing interleaved projection against a single post-hoc projection at the same selected order, at configuration A, interleaving is worse at 0 dB (-0.164 dB, interval $[-0.317, -0.066]$), indistinguishable from $+5$ to $+15$ dB, and better only at $+20$ dB ($+0.168$ dB, interval $[+0.102, +0.246]$), the six-point mean being $+0.003$ dB. Interleaving is therefore not required in this classical iteration.

Since placement is a wash, what the estimator contributes is not *where* the projection is applied but *whether* it is applied at all and at what order, decided from held-out pilots and using no oracle. The evidence is the abstention result above: the selector is what keeps $\Delta_{\text{HS}} \geq 0$, and forcing the projection past the crossing costs -0.075 dB with interval $[-0.186, -0.026]$. The order selector is the claim of this paper, not a mechanism explained on the way to one.

VI. LIMITATIONS AND CONCLUSION

An amplitude-only receiver’s information is block diagonal across elements, so aperture is usable only through cross-element structure; the exponential-sum structure of a ULA channel supplies it, with capacity $\lceil N/2 \rceil$. The constraint therefore helps only below a fraction of that capacity, the predicted sign reversal at small aperture is observed, and the governing coordinate $r_{\text{eff}}/r_{\text{max}}$ predicted the gain on two unseen channel distributions to within 0.13 and 0.315 dB.

Limitations. All results are simulation only, with no measured hardware and no recorded channel data. They cover geometric ULA channels of the form (2) at the array sizes, pilot counts and generators tested, and the cross-generator tests change the distribution but not the model family, so no claim is made beyond them. The gain magnitude does not collapse onto the proposed axis and its residual of 0.493 dB is unexplained, and ρ is not receiver-observable. Two of the three boundary crossings cannot be bracketed from the negative side, because abstention prevents it, and $\rho \approx 0.75$ is unreachable at $N = 64$ in any case, where the path generator fails above $L = 48$ ($\rho \approx 0.67$). We establish no fundamental bound: the references govern unbiased estimators and every estimator here is biased.

REFERENCES

- [1] M. Cui, Q. Zeng, and K. Huang, "Towards atomic MIMO receivers," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 3, pp. 659–673, 2025.
- [2] T. Gong *et al.*, "Rydberg atomic quantum receivers for classical wireless communication and sensing," *IEEE Wireless Commun.*, vol. 32, no. 5, pp. 90–100, 2025.
- [3] R. W. Gerchberg and W. O. Saxton, "A practical algorithm for the determination of phase from image and diffraction plane pictures," *Optik*, vol. 35, pp. 237–246, 1972.
- [4] J. Xiao, J. Wang, M. Zeng, H. Xu, X. Li, and A. Nallanathan, "Channel estimation for Rydberg atomic quantum receivers: unrolled phase retrieval from holographic snapshots," *IEEE Signal Process. Lett.*, vol. 33, pp. 1696–1700, 2026.
- [5] B. Xu, J. Zhang, Z. Chen, B. Cheng, Z. Liu, Y.-C. Wu, and B. Ai, "Channel estimation for Rydberg atomic receivers," *IEEE Wireless Commun. Lett.*, vol. 14, no. 9, pp. 2957–2961, 2025.
- [6] R. Balan, "The Fisher information matrix and the Cramér–Rao bound in a non-AWGN model for the phase retrieval problem," in *Proc. Int. Conf. Sampling Theory Appl. (SampTA)*, 2015.
- [7] Y. Chen and Y. Chi, "Robust spectral compressed sensing via structured matrix completion," *IEEE Trans. Inf. Theory*, vol. 60, no. 10, pp. 6576–6601, 2014.
- [8] K. H. Jin, D. Lee, and J. C. Ye, "A general framework for compressed sensing and parallel MRI using annihilating filter based low-rank Hankel matrix," *IEEE Trans. Comput. Imag.*, vol. 2, no. 4, pp. 480–495, 2016.
- [9] J. P. Haldar, "Low-rank modeling of local k -space neighborhoods (LORAKS) for constrained MRI," *IEEE Trans. Med. Imag.*, vol. 33, no. 3, pp. 668–681, 2014.
- [10] G. Tang, B. N. Bhaskar, P. Shah, and B. Recht, "Compressed sensing off the grid," *IEEE Trans. Inf. Theory*, vol. 59, no. 11, pp. 7465–7490, 2013.
- [11] J. A. Cadzow, "Signal enhancement—a composite property mapping algorithm," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 36, no. 1, pp. 49–62, 1988.
- [12] I. Markovsky, "Structured low-rank approximation and its applications," *Automatica*, vol. 44, no. 4, pp. 891–909, 2008.
- [13] O. Roy and M. Vetterli, "The effective rank: a measure of effective dimensionality," in *Proc. Eur. Signal Process. Conf. (EUSIPCO)*, 2007, pp. 606–610.
- [14] J. D. Gorman and A. O. Hero, "Lower bounds for parametric estimation with constraints," *IEEE Trans. Inf. Theory*, vol. 36, no. 6, pp. 1285–1301, 1990.
- [15] P. Stoica and B. C. Ng, "On the Cramér–Rao bound under parametric constraints," *IEEE Signal Process. Lett.*, vol. 5, no. 7, pp. 177–179, 1998.